Swami Keshvanand Institute of Technology Management & Gramothan, Jaipur

Course Title: CSUP120: C Programming Lab

Class: I Semester B.Tech **Exam Timings:** 3 Hours

Scheduled Lab per Week: 2 Hours **Max. Marks:** 100 (IA: 60, ETE: 40)

S.N.	**Computer Lab Assignment**
	Lab Assignment-1 (Basic C programs) *Structure of C Program, learn about the C library, Preprocessor directive, Input-Output statements, Data types, and Tokens in C Programming.*
P.1	Write a program to print "HELLO" message.
P.2	Write a program to add any two numbers.
P.3	Write a program to convert temperature from Celsius to Fahrenheit.
P.4	Write a program to find area and circumference of a circle.
	Lab Assignment-2 (Types of Operators) *Different types of operator Arithmetic, Relational, Logical, Increment/Decrement, Conditional, comma, Bitwise and sizeof() operator.*
P.5	Write a program to demonstrate the working of Arithmetic operators.
P.6	Write a program to demonstrate the working of Logical operators.
P.7	Write a program to demonstrate the working of Increment / Decrement operators.
P.8	Write a program to find largest among three numbers using conditional/ternary operator.
	Lab Assignment-3 (Conditional Statements) *Learn about various conditional statements of if-else, nested if-else , if-else-if (ladder) and switch case statements*
P.9	Write a program to check whether a number is Odd or Even.
P.10	Write a program to check a given character is Alphabet or Not.
P.11	Write a program using switch case statement to display from Monday to Sunday.
	Lab Assignment-4 (Looping Statements) *Learn about Iterative statements like while , do-while, for loop , continue and break statements*
P.12	Write a program to display first 10 Natural numbers and display their sum.
P.13	Write a program to print Table of a number using for loop.
P.14	Write a program to find whether number is Prime number or not.
P.15	Write a program to find Reverse of a given number.
P.16	Write a program to find Factorial of given number
	Lab Assignment-5 (Pattern based problem using Nested loop) Nested for loop, nested while loop, different kind of patterns.
P.17	Write a program to print half pyramid using * symbol.
P.18	Write a program to print half pyramid using numbers.
P.19	Write a program to print full pyramid using * symbol.

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	Lab Assignment-6 (Array) *Learn about array , types of array- one dimensional and two dimensional(matrix) operation on array like searching , sorting*
P.20	Write a program to calculate Sum and Average of 10 elements of given Array.
P.21	Write a program to find the maximum no in the Array.
P.22	Write a program for Addition of two matrices.
P.23	Write a program for Subtraction of two matrices.
P.24	Write a program for Multiplication of two matrices.
	Lab Assignment-7 (Functions and Recursion) *Learn about types of functions-user defined functions , parameter passing call by value and recursion process*
P.25	Write a program to find Prime numbers between a given ranges using function.
P.26	Write a program to find sum of n Natural Numbers using function.
P.27	Write a program to calculate Factorial of a number using recursion.
	Lab Assignment-8 (String and Character Array) *String declaration, initialization, different operations on string like copy, concatenate, reverse ,length, and comparison using string function and without using string function.*
P.28	Write a program to convert a string from lowercase to uppercase.
P.29	Write a program to find total number of alphabets, digits or special characters in a string.
	Lab Assignment-9 (Pointers) *Learn about pointers , arithmetic operations on pointers, pointer with array, pointer with string, pointer to pointer*
P.30	Write a program to create, initialize and use of pointers.
P.31	Write a program to perform all arithmetic operations using pointers.
P.32	Write a program to swap two numbers using call by references.
P.33	Write a program to use the concepts of double pointers (Pointers to Pointers).
	Lab Assignment-10 (Structure and Union) *Learn about structure ,declaration of structure, initialization,* ***nested structure****, structure pointer, array of structure , coping structure , passing structure to the user defined functions and* ***differentiate structure and union***
P.34	Write a program to read and print student details using structure pointers
P.35	Write a program to store Information and Display it using
P.36	Write a program to add two numbers using structure to a function Structure.
P.37	Write a program to create a array of structure for five students.

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	Lab Assignment-11 (File Handling) *File creation, types of file, modes of file (r, w, a, r+, w+, a+), different operation on file reading, writing, and appending, reading and writing structure data from/to the file.*
P.38	Write a program to read text from a file.
P.39	Write a program to write sentence in a file.
P.40	Write a program to find the total number of lines in a file.